



MSC MECHANICAL ENGINEERING

Imaging in Fluid Dynamics

Experiment 3 - PTV

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1 Abstract

This experiment investigates a quasi-two-dimensional (Q2D) electromagnetically driven flow using Particle Tracking Velocimetry (PTV). A conductive CuSO_4 solution is forced by the Lorentz interaction between an imposed magnetic field and an applied DC current. The image sequences were processed in PIVlab to obtain Eulerian velocity and vorticity fields for four current intensities: 0.01, 0.09, 0.25, and 0.49 A.

From the PTV analysis, the RMS velocity increases from approximately 5.3×10^{-3} to 3.5×10^{-2} m/s using boxcar smoothing, and from 5.4×10^{-3} to 3.6×10^{-2} m/s using Gaussian smoothing. The corresponding Reynolds numbers range from 187 to 1230 (boxcar) and from 188 to 1256 (Gaussian), following on average the expected scaling $Re \propto \sqrt{I}$. The characteristic flow time scale, $\tau_f = L_f/u_{\text{RMS}}$, decreases from roughly 6.5 s at 0.01 A to about 1.0 s at 0.49 A, showing how rapidly the large-scale flow evolves as the forcing increases.

Overall, the first three datasets reproduce the expected Lagrangian behavior: velocity PDFs remain close to a Gaussian core, acceleration PDFs show the correct intermittent structure, and the Lagrangian velocity autocorrelation exhibits the anticipated decorrelation and partial re-correlation at low Reynolds number. Only the strongest forcing case deviates from theoretical expectations, due to the severe reduction in trajectory length at high velocity, which prevents statistical convergence of the PDFs and autocorrelation functions.

Keywords: PTV, Q2D flow, Lorentz force, Lagrangian statistics, RMS velocity, Reynolds number, Flow time scale.

2 Introduction

The measurement of fluid velocity has evolved through several techniques, each improving how we observe and quantify flow motion. Early developments relied on Hot Wire Anemometry (HWA), where a small heated wire was placed directly in the flow. Variations in temperature or electrical resistance were used to infer the local velocity. Although effective, HWA is intrusive and only provides point measurements, which limits its use when spatial information is needed. As explained by Bruun [7], these limitations are mainly due to the intrusive nature of hot-wire anemometry.

In the 1970s, Laser Doppler Velocimetry (LDV) introduced a non-intrusive method based on the Doppler shift of light scattered by particles. LDV provides accurate velocity measurements without disturbing the flow, but it remains a point-based technique and requires precise optical alignment. Durst, Melling, and Whitelaw [8] describe in detail how these optical constraints and the focused nature of LDV limit its ability to capture spatial flow structures.

While PIV provides Eulerian velocity fields, its spatial averaging prevents accurate resolution of small-scale particle dynamics such as accelerations. Particle Tracking Velocimetry (PTV) overcomes this by following individual tracer particles in time, providing fully Lagrangian measurements.

PTV identifies particle centroids in each frame, with sub-pixel accuracy by applying a gaussian fitting, and links them across images to reconstruct trajectories. From these, velocities, accelerations, and Lagrangian statistics such as velocity autocorrelation, acceleration PDFs, and mean-squared displacement can be computed. Its performance depends on accurate particle detection, suitable tracer properties, and reliable tracking algorithms. PTV is particularly effective in quasi-two-dimensional (Q2D) flows, where limited out-of-plane motion allows long, continuous trajectories and improved

statistical convergence.

In this experiment, a turbulent Q2D flow is generated electromagnetically by driving a current through a conductive CuSO_4 solution above a checkerboard of permanent magnets. By varying the applied current, different Reynolds-number regimes are explored using PTV.

3 Theoretical Background

3.1 PTV Measurement Principle

PTV is a Lagrangian measurement technique, meaning that the fluid motion is described by following individual tracer particles as they move through the flow. Instead of estimating a spatially averaged displacement within interrogation windows, PTV identifies the position of each particle in every frame and reconstructs their trajectories by linking detections across time.

The measurement process begins by locating particle centroids in each image, with sub-pixel precision using Gaussian fitting. In the Lagrangian description, the flow is described by

$$\mathbf{X}(\mathbf{x}_0, t),$$

which gives the position at time t of the particle that was initially located at $\mathbf{x}_0$. The Lagrangian velocity and acceleration of this particle are then defined as

$$\mathbf{u}(\mathbf{X}(\mathbf{x}_0, t), t) = \frac{\partial \mathbf{X}}{\partial t}(\mathbf{x}_0, t), \quad \mathbf{a}(\mathbf{X}(\mathbf{x}_0, t), t) = \frac{\partial^2 \mathbf{X}}{\partial t^2}(\mathbf{x}_0, t).$$

PTV provides access to detailed Lagrangian statistics, including velocity autocorrelation, acceleration probability distributions, and mean-squared displacement. In quasi-two-dimensional (Q2D) flows, out-of-plane motion is minimal, enabling long trajectories and improving statistical convergence. Figure 1 illustrates the principle of particle identification and trajectory reconstruction used in PTV.

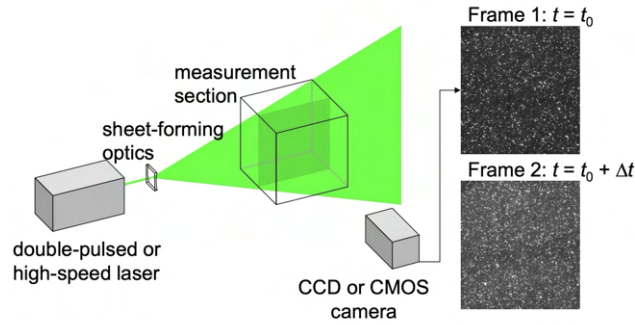


Figure 1: System of pulse laser to capture frames at different instant.

3.2 Lagrangian Velocity Autocorrelation Function

The Lagrangian concept can be compared to a fisherman sitting in a small boat that drifts with the river. Instead of standing on the shore and observing the water flowing past, the fisherman now moves together with the current and experiences how his own velocity evolves over time. In the same way, PTV follows individual particles as they are carried by the flow, providing a trajectory-based

description of the motion rather than a field defined at fixed spatial locations.

In the Lagrangian framework, the flow is described by following individual particles along their trajectories. Instead of evaluating how velocity varies at fixed spatial locations, we examine how a particle remembers its past motion as it is advected by the flow. This memory is quantified using the Lagrangian velocity autocorrelation function (VACF).

The normalized Lagrangian VACF for a time lag t is

$$\Gamma_L(t) = \frac{\langle \mathbf{u}_i(t_0) \cdot \mathbf{u}_i(t_0 + t) \rangle_{i,t_0}}{\langle \mathbf{u}_i(t_0) \cdot \mathbf{u}_i(t_0) \rangle_{i,t_0}},$$

where the average is taken over all particles i and over all possible starting times t_0 . This correlation measures how long a particle retains information about its initial velocity.

The decay of $\Gamma_L(t)$ reflects how long a particle remembers its initial velocity, as described in the PTV manual. A characteristic decay time τ may be defined as the time at which $\Gamma_L(t)$ reaches $1/e$ of its initial value. The manual also emphasizes that all possible time origins t_0 along each trajectory must be used when computing $\Gamma_L(t)$, since relying only on the initial frame of each trajectory leads to poor statistical convergence and noisy correlation curves.

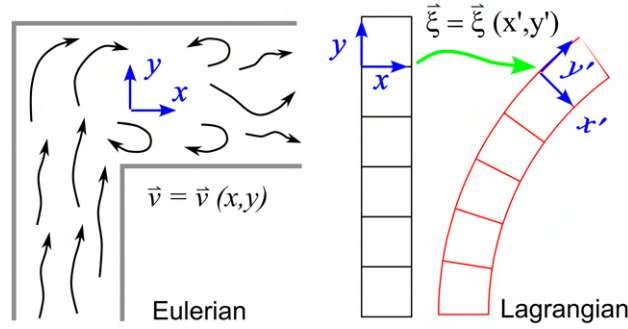


Figure 2: Eulerian perspective of fluid velocity versus Lagrangian depiction of strain [5].

3.3 Physical Variables

3.3.1 Lorentz Force

The electromagnetic forcing applied in this experiment arises from the Lorentz force per unit mass:

$$\mathbf{f}_b = \frac{\mathbf{J} \times \mathbf{B}}{\rho},$$

where $\mathbf{J}$ is the electric current density, $\mathbf{B}$ the magnetic field, and ρ the fluid density. This interaction generates alternating counter-rotating vortices above the magnet array and produces the quasi-two-dimensional flow observed in the experiment.

3.3.2 Reynolds Number and Average Velocity

The characteristic velocity used in this work is the root-mean-square velocity, u_{RMS} . For each frame, PIVlab provides the magnitude of the velocity vectors over all interrogation windows. An RMS value is computed for every frame, and the global u_{RMS} is obtained by taking the RMS of these frame-by-frame values. This quantity represents an overall measure of the velocity intensity in the flow.

From theory, the characteristic flow velocity in this type of electromagnetically driven quasi-two-dimensional system is expected to scale as

$$U \approx \left(\frac{NBIL_f^2}{V\rho} \right)^{1/2},$$

which leads directly to the relation

$$Re = \frac{UL_f}{\nu} \approx \frac{L_f^2}{\nu} \left(\frac{NBI}{\rho V} \right)^{1/2}.$$

This shows that the Reynolds number is expected to follow the scaling

$$Re \propto \sqrt{I}.$$

In the Lagrangian framework, the RMS velocity is obtained by averaging the squared particle velocities over all trajectories and all time instants:

$$Re = \frac{u_{\text{RMS}} L_f}{\nu}, \quad \text{with} \quad u_{\text{RMS}} = \langle \mathbf{u}_i(t) \cdot \mathbf{u}_i(t) \rangle_{i,t}^{1/2},$$

where the average is taken over all particles i and all time samples along their trajectories, $L_f = 35$ mm is the magnet spacing and $\nu \approx 1$ mm²/s is the kinematic viscosity of the CuSO₄ solution.

3.4 Mean-Squared Displacement

The Mean-Squared Displacement (MSD) describes how far a particle moves away from its initial position over a time interval t . For a particle trajectory $\mathbf{x}_i(t)$, it is defined as

$$\langle r^2(t) \rangle = \left\langle |\mathbf{x}_i(t_0 + t) - \mathbf{x}_i(t_0)|^2 \right\rangle_{i,t_0}.$$

At short times, particles typically move ballistically, leading to a quadratic growth $\langle r^2(t) \rangle \propto t^2$. As velocity correlations decay, the motion becomes diffusive and the MSD grows linearly in time, $\langle r^2(t) \rangle \propto t$. This transition from ballistic to diffusive behavior reflects how particle motion evolves in a turbulent flow.

3.5 Nearest Neighbors and Smoothing

PTV processing consists first in locating tracer particles in each frame by identifying intensity maxima, since tracers appear brighter than the background. Sub-pixel accuracy is obtained by fitting a 2D Gaussian to the local intensity map. Once particles are detected, trajectories are reconstructed using the nearest-neighbor method, where each particle in frame k is associated with the closest particle in frame $k + 1$. Repeating this procedure over all frames yields individual Lagrangian paths. To obtain reliable velocities and accelerations from these trajectories, the raw positions must be smoothed. Gaussian smoothing replaces each position by a weighted average of surrounding points using the kernel

$$g(\tau) = \frac{1}{\sqrt{\pi}\omega^2} \exp\left(-\frac{\tau^2}{\omega^2}\right),$$

where points closer in time receive greater weight. Because differentiation commutes with convolution, smoothed velocities and accelerations are obtained by convolving the trajectory with the derivatives of this kernel. The width ω is chosen just large enough to suppress noise while preserving genuine flow dynamics.

A simpler alternative is the boxcar filter, which assigns equal weight to all points within a fixed temporal window. This method effectively removes high-frequency fluctuations but offers less temporal precision than the Gaussian kernel.

4 Experimental Setup

The measurement system consists of an experimental flow tray with internal dimensions of $300 \text{ mm} \times 320 \text{ mm}$ placed under a high-speed camera mounted on a rigid support. At both ends of the tank, two copper electrodes are installed and connected to a DC power supply, allowing an electric current to pass through the working fluid. Beneath the flow tray, an 8×8 magnetic checkerboard made of neodymium magnets generates an alternating magnetic field. The center-to-center spacing between magnets is $L_f = 35 \text{ mm}$, which sets the characteristic forcing length scale of the flow.

Two blue LED panels positioned above the tank provide uniform illumination of the flow. The tank is filled with a 10% CuSO_4 solution (density $\rho_f = 1080 \text{ kg/m}^3$) forming a $h = 5 \text{ mm}$ deep fluid layer. Fluorescent green polyethylene microspheres with radii in the range $75\text{--}90 \text{ }\mu\text{m}$ and density $\rho_p = 1000 \text{ kg/m}^3$ are added as tracer particles. Their size and near-neutral buoyancy allow them to follow the flow motion accurately.

Image acquisition is performed using a Phantom VEO 640 high-speed CMOS camera with a 1280×1280 pixel sensor capable of recording up to 3260 frames per second. The camera is operated through the Phantom Camera Control software, which provides real-time monitoring and image management. A checkerboard calibration plate matching the fluid layer thickness is used prior to recording to convert pixel dimensions into physical units.

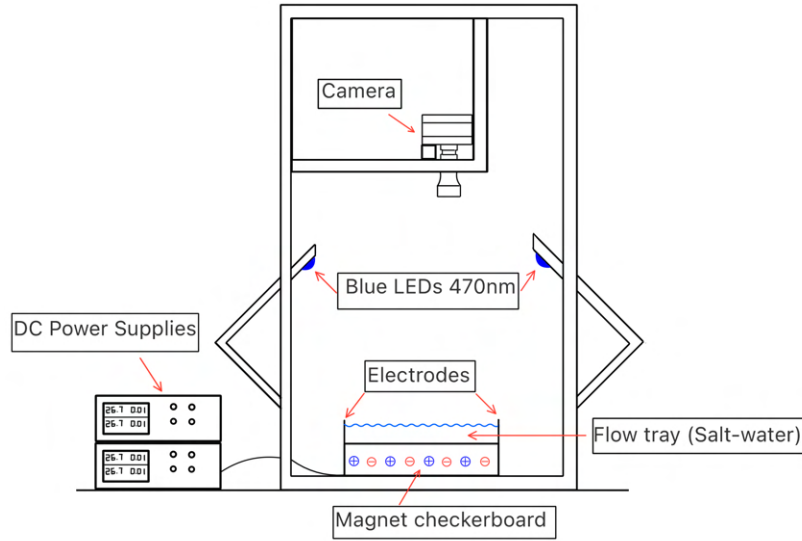


Figure 3: Experimental setup of the Q2D experiment.

5 Methods and Data Processing

5.1 Experimental Procedure

The experiment begins by preparing the CuSO_4 solution used as the working fluid. A 10% mass concentration is obtained by dissolving $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in water and mixing the solution with a magnetic stirrer until fully homogeneous. The prepared solution is then poured into the flow tray.

Particle seeding for PTV: In contrast to PIV, where a relatively dense distribution of tracers is required inside each interrogation window, PTV demands a much lower seeding density. Only

a small, visually estimated amount of particles is added so that individual tracers remain clearly separated in the images. This ensures that each particle can be uniquely identified and tracked from frame to frame without ambiguity. A drop of dish soap is used to reduce surface tension and prevent particle clustering. The fluid is gently mixed before each recording to maintain a uniform distribution of tracers.

Practical steps: The camera is configured with the selected frame rate and exposure time. After waiting briefly for the flow to become fully steady, image sequences are recorded for each applied current. This procedure is repeated for all experimental conditions.

5.2 Calibration and Data Processing

In the last experiment, a spatial calibration was performed using a checkerboard target placed at the height of the fluid layer. The pixel-to-meter conversion factor was obtained as

$$1 \text{ px} = 8.575 \times 10^{-5} \text{ m.}$$

This factor was used consistently through all datasets to convert the PTV velocities from pixels per frame to physical units.

To ensure reliable tracking of each particle, several processing parameters had to be adjusted for every experimental condition. As the applied current increases, the flow velocity becomes larger and particles move farther between frames, so both the number of recorded frames and the search radius were adapted so that inter-frame displacements remained within a trackable range. In addition, thresholds on particle intensity and acceptable particle radius were imposed during detection in order to filter out noise and non-physical blobs. By retaining only particles within well-defined intensity and size ranges, the tracking remained stable and trajectories could be reconstructed without mismatches.

Current (A)	Search radius (px)	FPS	Intensity range	Particle radius (px)
0.01	5	50	[190 – 1600]	[1.6 – 2.6]
0.09	5	75	[200 – 1700]	[1.6 – 2.6]
0.25	5	100	[200 – 1400]	[1.6 – 2.6]
0.49	7	150	[160 – 1200]	[1.6 – 2.6]

Table 1: PTV tracking parameters and particle detection thresholds for each applied current.

After the tracking stage, the particle trajectories must be smoothed by selecting an appropriate kernel width. The choice of this parameter is crucial, since an overly narrow kernel fails to remove position noise, whereas an overly wide kernel suppresses genuine flow dynamics. According to theory, one should select the smallest kernel width beyond which the acceleration variance begins to decay exponentially. In this experiment, we follow the procedure proposed by Tropea [1], where the optimal kernel width is identified by examining the evolution of acceleration variance as a function of smoothing strength as we can see on the figure below.

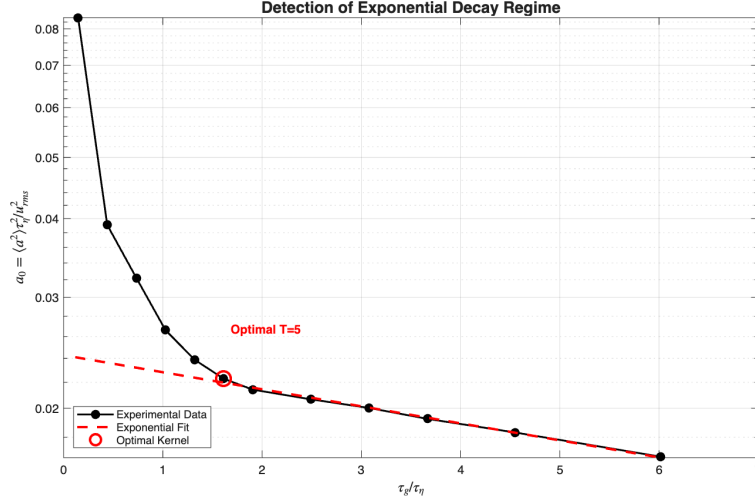


Figure 4: Normalized acceleration as a function of kernel width normalized by τ_η for the 0.25A regime.

6 Results

6.1 Trajectories and Eulerian Conversion

The particle trajectories, along with the reconstructed velocity and vorticity fields, are presented in Section 9. These visualizations allow us to observe how the flow structure evolves as the applied current increases, and how the Lagrangian particle motion reflects the transition from an ordered regime to a more turbulent one.

Trajectory Visualization

At low current, the trajectories are long, smooth and well organized, reflecting coherent particle motion in a laminar or weakly perturbed regime. As the Reynolds number increases, the trajectories become progressively shorter and more irregular. This behaviour is consistent with what Ouellette have shown in their study of particle tracking in turbulent flows, namely that increasing fluctuations reduce temporal coherence and rapidly fragment trajectories [2]. In the highest current case, Figure 18 clearly demonstrates that each particle can only be tracked over a very short frame length, producing fragmented paths that visually resemble "confetti".

Discussion

Although the parameters $bplo$, $bphi$ and the search radius strongly influence the recovered frame length since inappropriate values can drastically reduce the ability to track particles as described by Maas in their early PTV work [3], extensive testing showed that even with optimised values the trajectory length cannot be extended further. Once the parameters lie within a physically reasonable range, additional tuning does not increase the frame length. The severe reduction observed at high current therefore reflects the intrinsic physics of the flow, in particular the increased velocity, rapid decorrelation and enhanced out of plane motion at high Reynolds number.

Regarding the Eulerian fields, the agreement between PIV and PTV is good for the lowest current. In this regime the flow is nearly laminar and sufficiently steady for both methods to capture similar velocity and vorticity patterns. At higher forcing, however, differences appear between the PIV and

PTV fields. This behavior is expected because PIVlab reconstructs an instantaneous Eulerian field from a single image pair while the Eulerian conversion applied to PTV data averages particle velocities onto a spatial grid using many trajectories and times. This averaging smooths rapid temporal variations and blends information from particles passing through a location at different instants. As the flow becomes increasingly unsteady at high Reynolds numbers, the two representations naturally diverge except in the laminar case where the flow remains nearly stationary.

Uncertainties and Performance Estimation

The ability to track particles reliably also depends on the temporal resolution. The performance criterion introduced in the lecture is $\xi = u\Delta t/d_{NN}$, which must remain below 0.1 to avoid mismatches in nearest neighbor tracking. The inter particle spacing d_{NN} depends on the seeding density. Santiago and co authors have shown that excessive seeding reduces this spacing and significantly increases the chance of tracking errors [4]. Assuming approximately ten particles per interrogation window gives $d_{NN} \approx 2.5$ mm. Using the smoothed value $u_{RMS} \approx 35$ mm s⁻¹ at 0.49 A leads to

$$(\Delta t)^{-1} > \frac{u_{RMS}}{0.1 d_{NN}} \approx \frac{35}{0.25} \approx 140 \text{ fps.}$$

Our acquisition framerate of 150 fps satisfies this requirement but only with limited margin. Two additional sources of uncertainty must be considered. First, the real seeding density may have been higher than assumed, which would reduce d_{NN} and increase the required framerate. Second, the smoothed velocity likely underestimates peak particle velocities, implying that the true value of ξ may be larger. A safer acquisition rate would therefore have been around 200 fps. However, such a rate would have produced considerably larger datasets and significantly increased storage and computation time. The reduced frame length observed at high current is thus primarily a physical limitation of the flow rather than a sampling deficiency.

6.2 Reynolds Numbers

In the previous section, the PIV analysis confirmed the theoretical scaling $Re \propto \sqrt{I}$ for electromagnetically driven quasi-two-dimensional flows. The experimentally measured Reynolds numbers followed this relation closely, indicating that the characteristic velocity indeed grows with the square root of the applied current. Having established this behavior using Eulerian measurements, and as discussed in Section 3.3.2, we now turn to the Lagrangian results obtained from PTV. By computing the RMS velocity from particle trajectories and applying the same definition $Re = u_{RMS}L_f/\nu$, we can assess whether the Lagrangian measurements produce Reynolds numbers consistent with the PIV results and with the expected theoretical trend. The values of u_{RMS} and Re obtained using both boxcar and Gaussian smoothing are summarized below.

Table 2: Comparison of PTV results using Boxcar and Gaussian smoothing.

Current (A)	Boxcar		Gaussian	
	u_{RMS} (m/s)	Re	u_{RMS} (m/s)	Re
0.01	0.005 346	187.12	0.005 380	188.29
0.09	0.016 964	593.73	0.017 027	595.96
0.25	0.024 833	869.16	0.024 999	874.97
0.49	0.035 135	1 229.74	0.035 890	1 256.15

For completeness, the characteristic flow time scale can be estimated as

$$\tau_f = \frac{L_f}{u_{RMS}},$$

which represents the time required for a fluid structure to travel across one forcing cell. Using $L_f = 35$ mm and the measured RMS velocities for the four forcing levels, we obtain

$$\tau_f^{0.01A} \approx 7.9 \text{ s}, \quad \tau_f^{0.09A} \approx 2.1 \text{ s}, \quad \tau_f^{0.25A} \approx 1.4 \text{ s}, \quad \tau_f^{0.49A} \approx 1.0 \text{ s}.$$

These values quantify how rapidly the large-scale flow evolves and help assess the temporal resolution needed to resolve particle motion reliably.

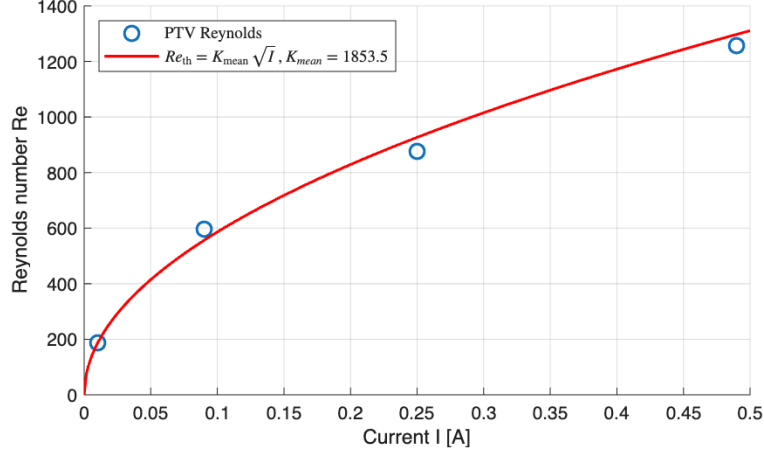


Figure 5: Evolution of the experimental Reynolds numbers compared to PTV theoretical scaling

Discussion: We observe that the Reynolds numbers obtained from PTV lie below the theoretical curve derived from the PIV analysis for the 2 last values using $K \approx 2049$. The deviation is most pronounced at higher currents, where the largest particle displacements and the stronger turbulence make Lagrangian measurements more sensitive to tracking errors and smoothing. Nevertheless, the four PTV data points still follow a clear $\sqrt{I}$ trend. By fitting the relation $Re = K_{PTV}\sqrt{I}$ to the PTV results, we obtain a new proportionality factor $K_{PTV} \approx 1853$, confirming that the Lagrangian Reynolds number continues to scale with the square root of the applied current.

The differences between Boxcar and Gaussian smoothing remain very small, indicating that both filters suppress high-frequency noise to a comparable degree. However, the Gaussian-smoothing results are consistently slightly closer to the theoretical trend. This is expected since Gaussian convolution provides a weighted average that preserves temporal coherence more effectively than the uniform Boxcar filter, as discussed in Tropea’s treatment of smoothing kernels [1] and in the PTV manual. By giving more weight to points near the evaluation time, the Gaussian filter yields more accurate estimates of velocity and acceleration, which in turn produces Reynolds numbers that better reflect the underlying flow physics.

The table below compares the RMS velocities and Reynolds numbers measured using PIV and PTV for the same forcing conditions.

Table 3: Comparison between Eulerian (PIV) and Lagrangian (PTV) measurements of u_{RMS} and Re .

Current (A)	PIV		PTV (Gaussian)	
	u_{RMS} (m/s)	Re	u_{RMS} (m/s)	Re
0.01	0.004 412	154.408	0.005 380	188.29
0.09	0.017 100	598.496	0.017 027	595.96
0.25	0.029 382	1 028.377	0.024 999	874.97
0.49	0.041 324	1 446.355	0.035 890	1 256.15

6.3 Velocity and Acceleration Probability Distribution

For the PTV analysis, the probability density functions of velocity and acceleration are computed directly from the smoothed particle tracks. For each dataset and each current, the Lagrangian velocity components (v_x, v_y) and accelerations (a_x, a_y) of all particles and all time steps are gathered into a single ensemble. Fluctuations are obtained by subtracting the mean value of each component, and they are normalized by their respective standard deviations, yielding dimensionless variables

$$u'_{\text{norm}} = \frac{u' - \langle u' \rangle}{\sigma_{u'}}, \quad a'_{\text{norm}} = \frac{a' - \langle a' \rangle}{\sigma_{a'}}.$$

The normalized samples from both components are then combined, and smooth PDFs are estimated using a kernel density estimator over a common symmetric range. The resulting distributions are finally compared to a standard Gaussian $\mathcal{N}(0, 1)$, using a logarithmic scale on the vertical axis to highlight the behavior of the tails for the different flow conditions and smoothing methods.

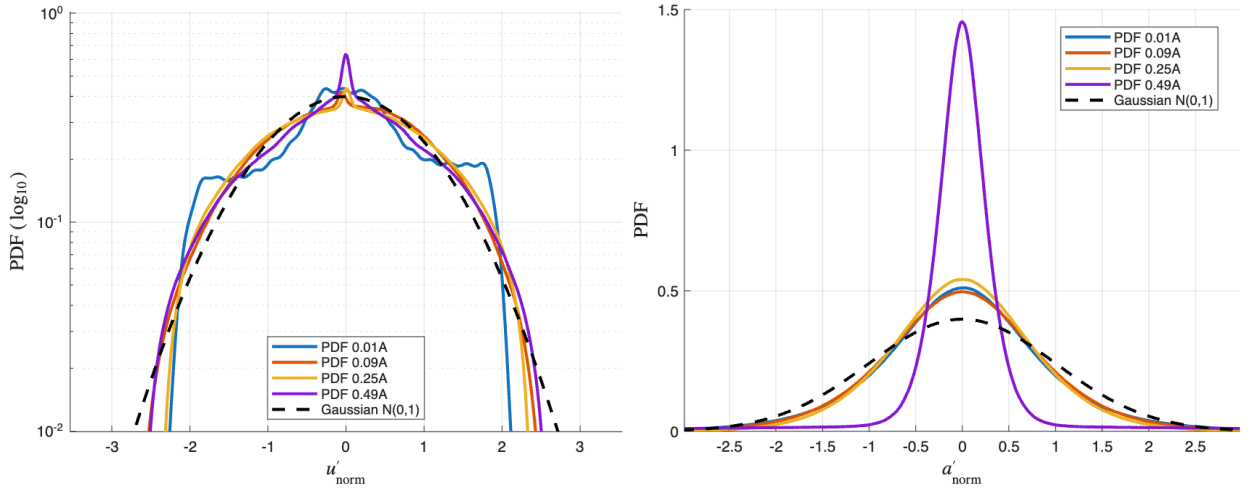


Figure 6: Probability distribution of the velocity magnitude and the normalized velocity fluctuations.

Discussion: In Lagrangian turbulence, the normalized PDF of velocity fluctuations is expected to approach a Gaussian shape as the Reynolds number increases, while the acceleration PDF is known to remain strongly non-Gaussian with a sharp peak at zero and heavy tails, as demonstrated by La Porta et al. in their landmark experiment [6]. The first three datasets follow these tendencies reasonably well, showing a clear symmetric peak in the velocity PDFs and broader acceleration distributions. However, the fourth dataset deviates strongly from these expected behaviors. As discussed previously, the trajectories at high current are extremely short, which severely limits the number of independent samples and prevents the PDFs from converging. These tracking limitations, rather than a physical change in the flow, explain the distorted shapes.

6.4 Lagrangian Velocity Autocorrelation and Mean Square Displacement

As discussed in Section 3.2, the Lagrangian velocity autocorrelation function quantifies how long a particle retains memory of its initial velocity while it is carried by the flow. For each trajectory, the inner product $\mathbf{u}(t_0) \cdot \mathbf{u}(t_0 + t)$ is computed for all possible starting times t_0 , and the correlation is normalized by $\mathbf{u}(t_0)^2$. Averaging over all particles and all time origins produces the Lagrangian correlation $\Gamma_L(t)$, whose decay characterizes the temporal coherence of the flow.

In parallel, the mean square displacement (MSD) describes how far particles move away from their initial position. The MSD typically exhibits an initial ballistic regime proportional to t^2 , followed by a linear diffusive regime as velocity correlations decay. Together with $\Gamma_L(t)$, the MSD provides a complementary view of how turbulence influences particle dispersion.

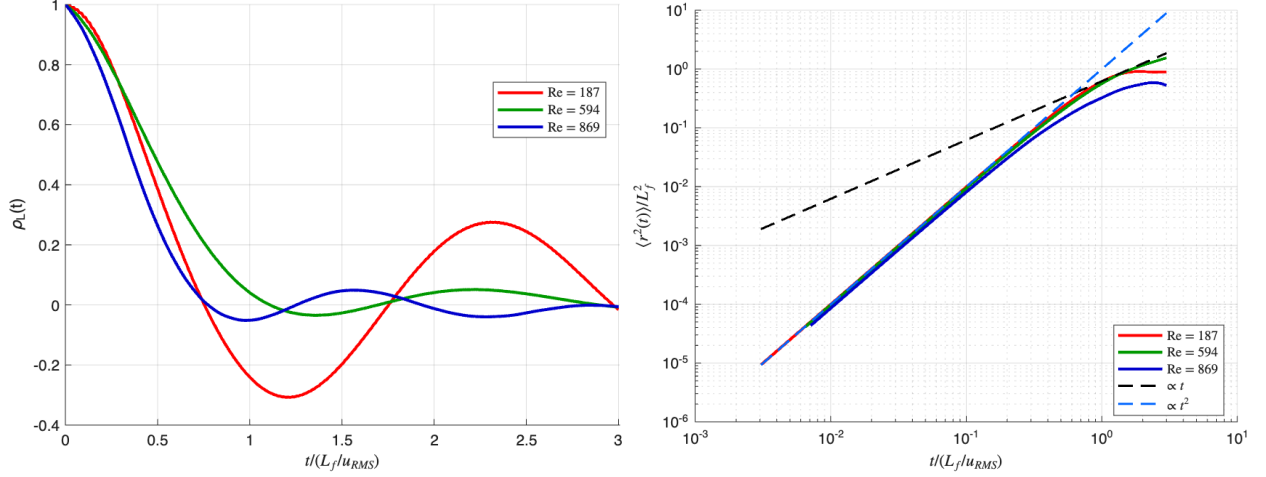


Figure 7: VACF and MSD evolution

Discussion: For the lowest current, the Lagrangian velocity autocorrelation does not decrease monotonically. After an initial drop, the curve increases again, showing that particle velocities regain some correlation at later times. This behaviour reflects the presence of coherent flow structures that persist long enough for particles to experience similar motions after a certain delay. As the Reynolds number increases, this re-correlation gradually disappears and the decay becomes much faster, indicating that velocity fluctuations lose coherence more rapidly. In the highest forcing case, the correlation falls almost immediately and shows no recovery, which is characteristic of strongly turbulent motion.

The mean square displacement exhibits the corresponding behaviour. At short times, the MSD grows proportionally to t^2 , representing the ballistic regime. As the velocity correlation decays, the MSD transitions to a linear growth in time, characteristic of diffusive particle spreading. This transition occurs more abruptly at higher currents because the decorrelation happens more quickly, consistent with what is observed in the autocorrelation curves.

7 Conclusion

This second part of the experiment applied Particle Tracking Velocimetry to the same quasi-two-dimensional electromagnetically driven flow, with the aim of characterizing its Lagrangian dynamics across four forcing levels. From the particle trajectories, we extracted the RMS velocities, Reynolds numbers, velocity and acceleration probability distributions, Lagrangian velocity autocorrelation functions, and mean square displacements.

While the general trend $Re \propto \sqrt{I}$ is still observed, the quantitative agreement with the PIV-based scaling deteriorates at high forcing. This discrepancy originates from the rapid loss of trackability at large Reynolds number, where particles can only be followed over very short frame lengths. As a consequence, some of the Lagrangian statistics do not converge toward their theoretical behavior. Similarly, For the 4th one, the Lagrangian velocity autocorrelation decays too abruptly to capture the

expected dynamical features of the turbulent flow, as illustrated in the Section 9. These limitations are linked to insufficient temporal resolution and the reduced number of statistically independent samples available in the most strongly forced case.

It would therefore be highly valuable to repeat the experiment with a higher acquisition frequency, for example 200 fps or above, in order to ensure smaller inter-frame displacements and longer trajectories. Comparing the current datasets with higher-fidelity PTV measurements would help clarify how much of the deviation comes from physical effects and how much is due to sampling limitations. Additional experiments with improved seeding control or multi-camera setups could also provide more reliable acceleration statistics and a clearer view of Lagrangian dispersion in Q2D turbulence. Such extensions would allow a more robust comparison between Eulerian and Lagrangian descriptions of the same flow and offer deeper insights into the mechanisms governing the transition from coherent to turbulent dynamics in electromagnetically driven layers.

8 References

References

- [1] Tropea, C. (2007). *Optical Measurement Techniques*. In: Tropea, C., Yarin, A. L., & Foss, J. F. (eds.), *Springer Handbook of Experimental Fluid Mechanics*. Springer.
- [2] Ouellette, N. T., Xu, H., & Bodenschatz, E. (2006). *A quantitative study of three-dimensional Lagrangian particle tracking algorithms*. Experiments in Fluids.
- [3] Maas, H. G., Gruen, A., & Papantoniou, D. (1993). *Particle tracking velocimetry in three-dimensional flows*. Experiments in Fluids.
- [4] Santiago, J. G., Wereley, S. T., Meinhart, C. D., Beebe, D. J., & Adrian, R. J. (1998). *A particle tracking method for experimental microfluidic flows*. Experiments in Fluids.
- [5] Wikipedia contributors. *Lagrangian and Eulerian specification of the flow field*. Wikipedia
- [6] La Porta, A., Voth, G. A., Crawford, A. M., Alexander, J., & Bodenschatz, E. (2001). *Fluid particle accelerations in fully developed turbulence*. Nature.
- [7] Bruun H. H. (1995). *Hot-Wire Anemometry: Principles and Signal Analysis*. Oxford University Press.
- [8] Durst F., Melling A., & Whitelaw J. H. (1976). *Principles and Practice of Laser-Doppler Anemometry*. Academic Press.

9 Appendix

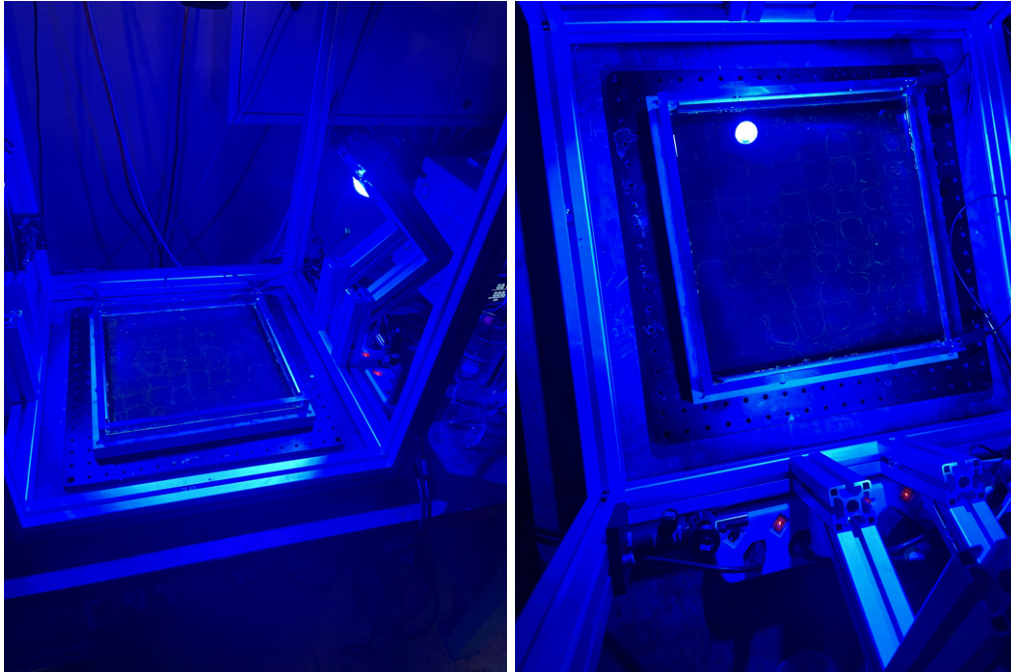


Figure 8: Experimental setup photo.

First mode: 0.01A

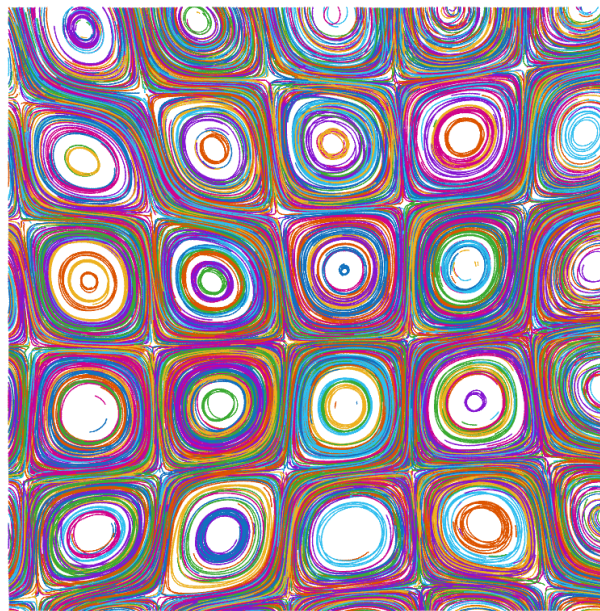


Figure 9: PTV Particle tracing

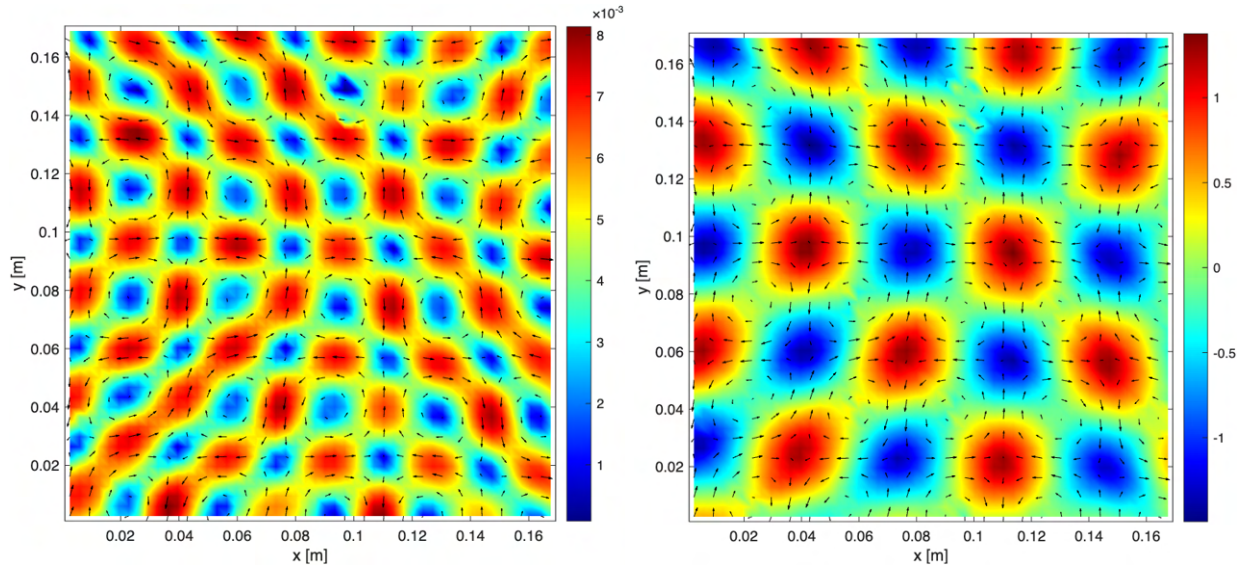


Figure 10: Eulerian conversion map of Vorticity and Magnitude

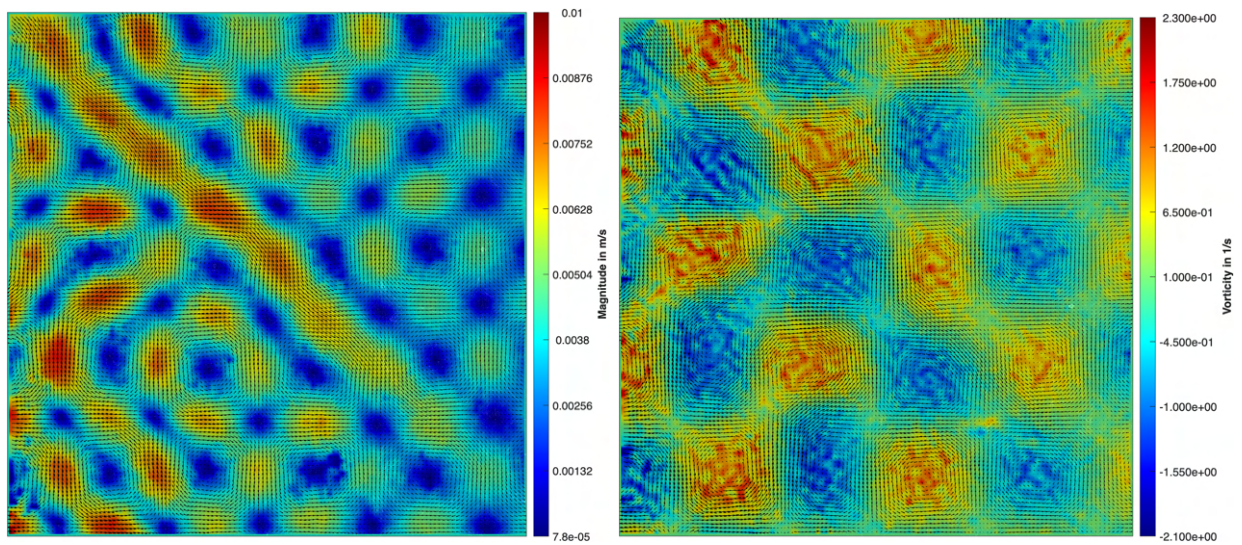


Figure 11: PIV map of Vorticity and Magnitude

Second mode: 0.09A

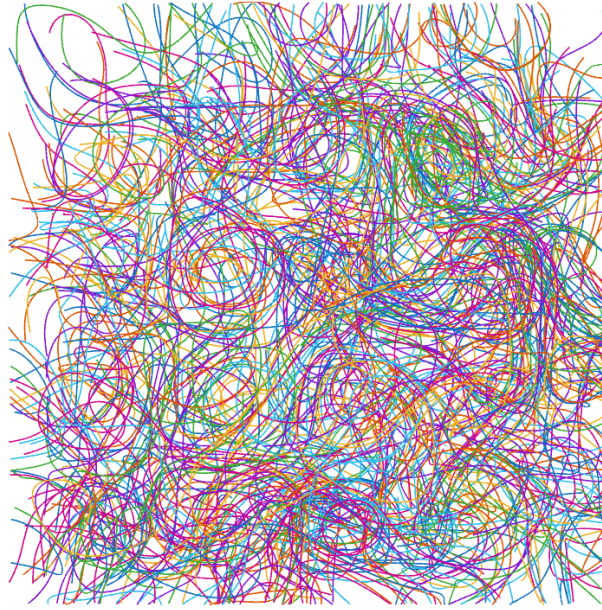


Figure 12: PTV Particle tracing

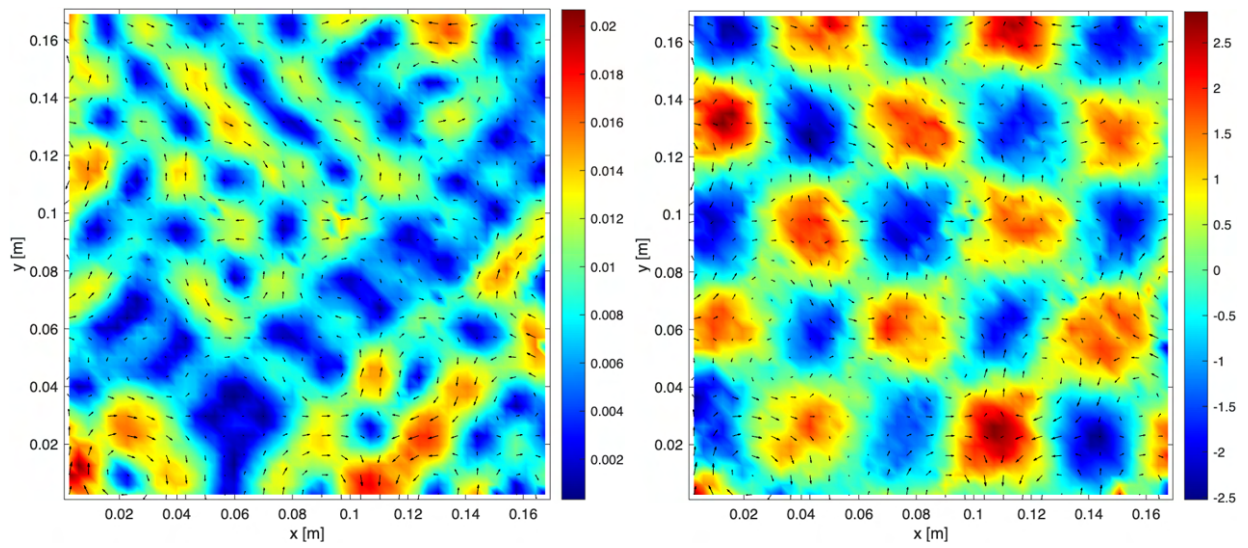


Figure 13: Eulerian conversion map of Vorticity and Magnitude

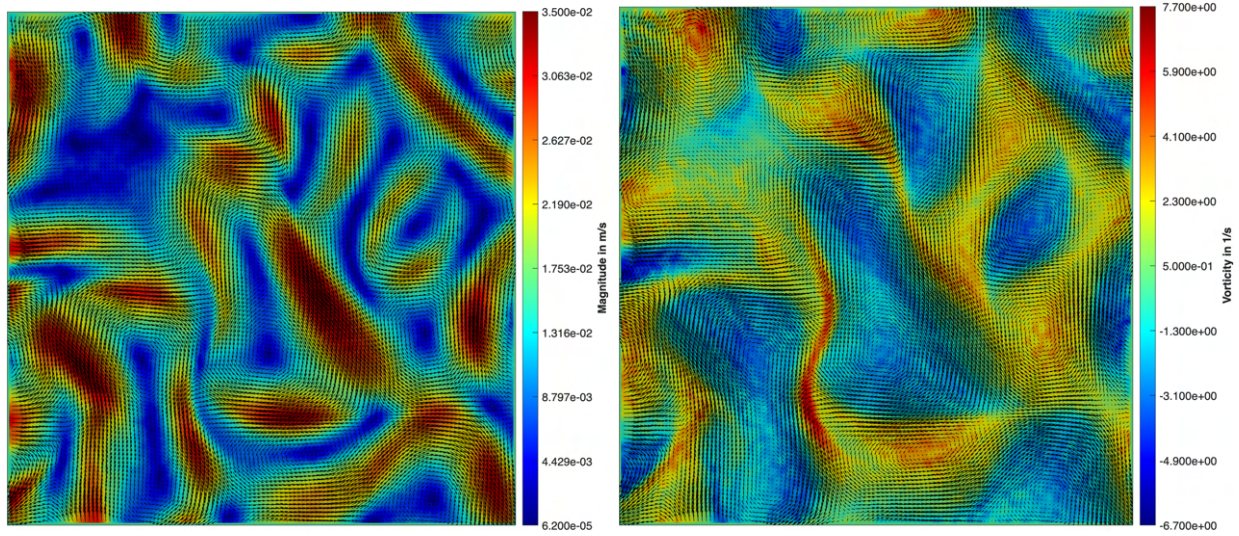


Figure 14: PIV map of Vorticity and Magnitude for 0.09A

Third mode: 0.25A

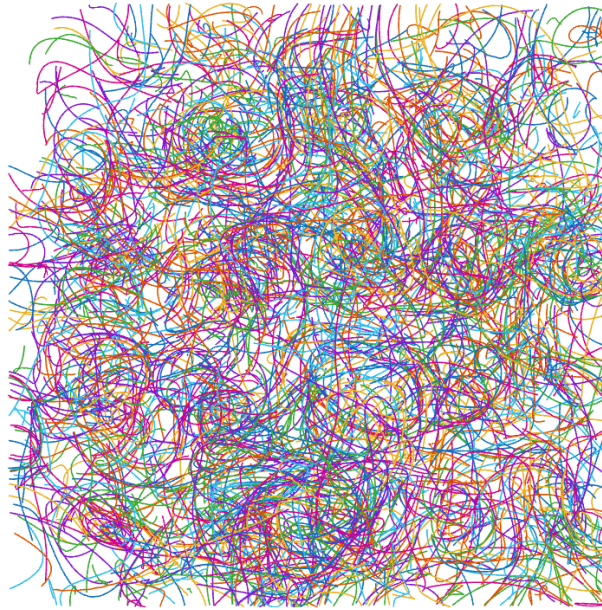


Figure 15: PTV Particle tracing

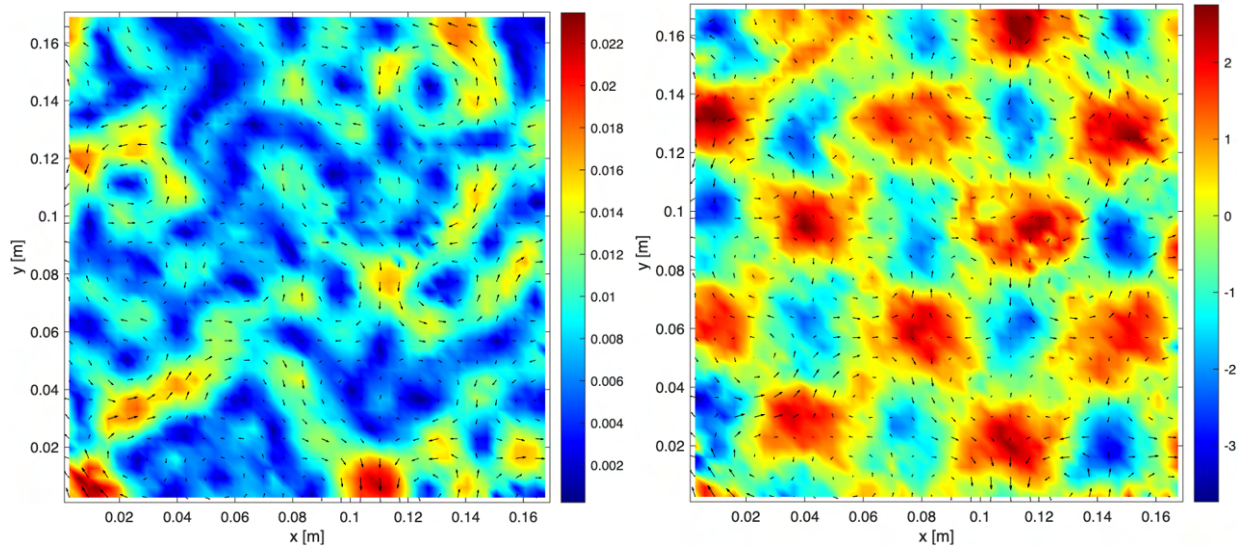


Figure 16: Eulerian conversion map of Vorticity and Magnitude

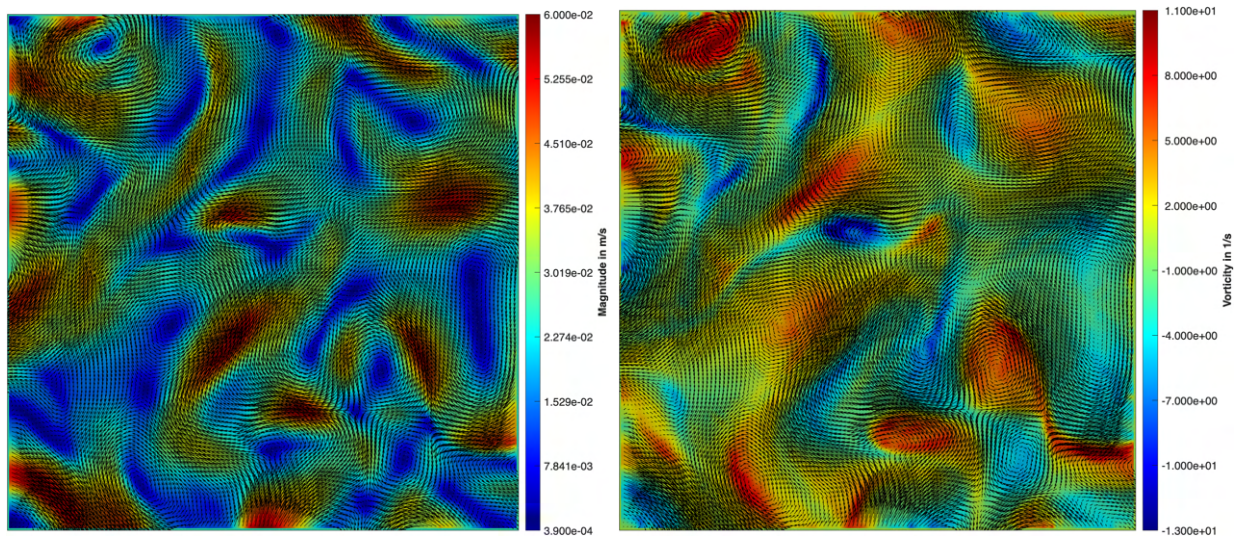


Figure 17: PIV map of Vorticity and Magnitude for 0.25A

Fourth mode: 0.49A

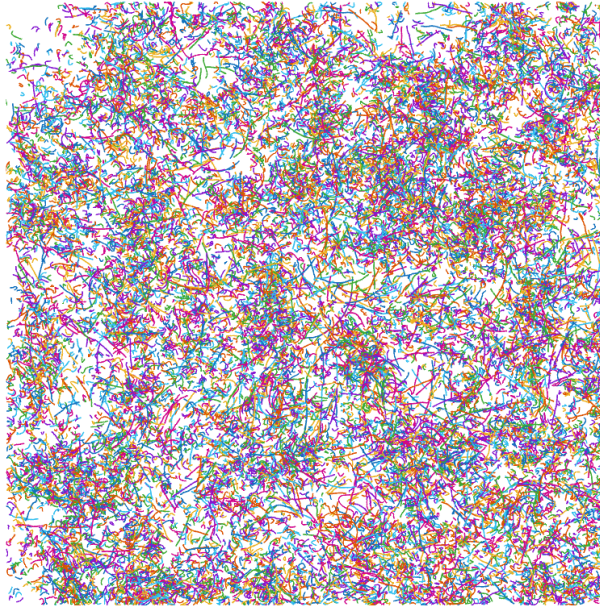


Figure 18: PTV Particle tracing

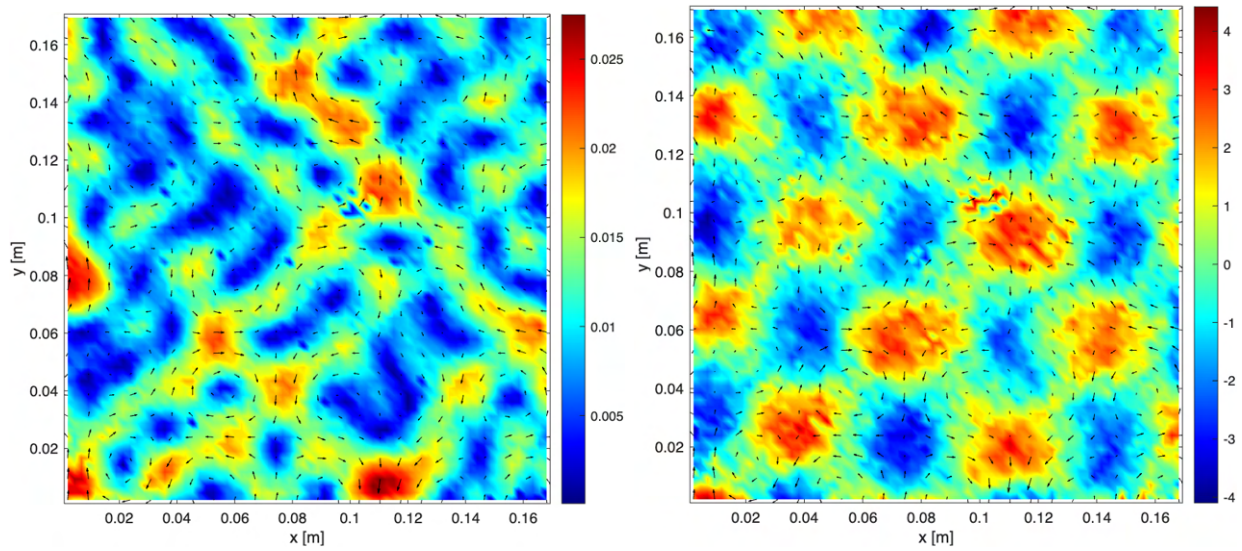


Figure 19: Eulerian conversion map of Vorticity and Magnitude

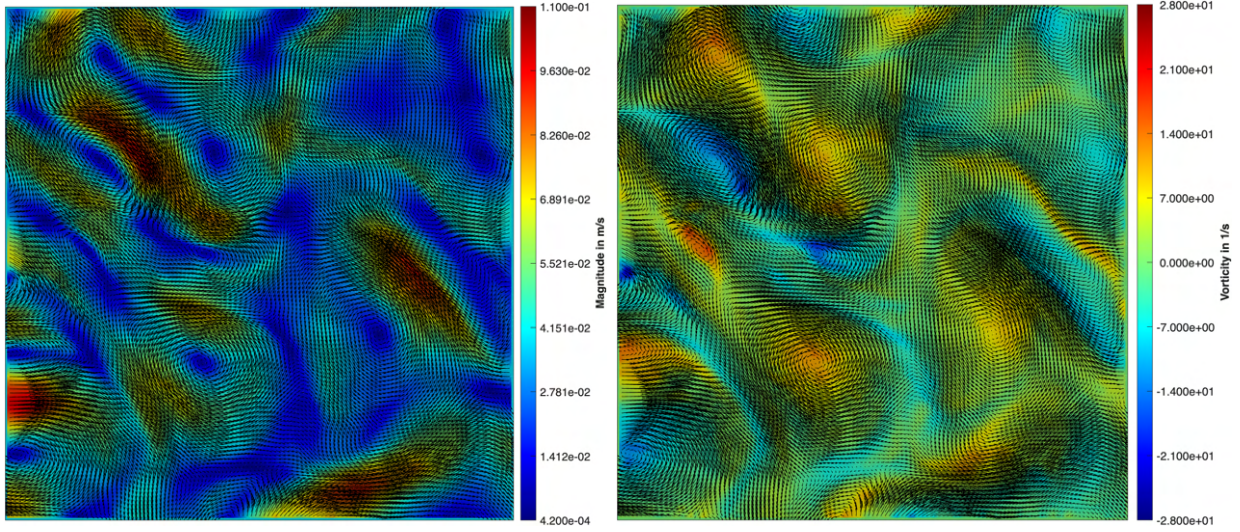


Figure 20: PIV map of Vorticity and Magnitude for 0.49A

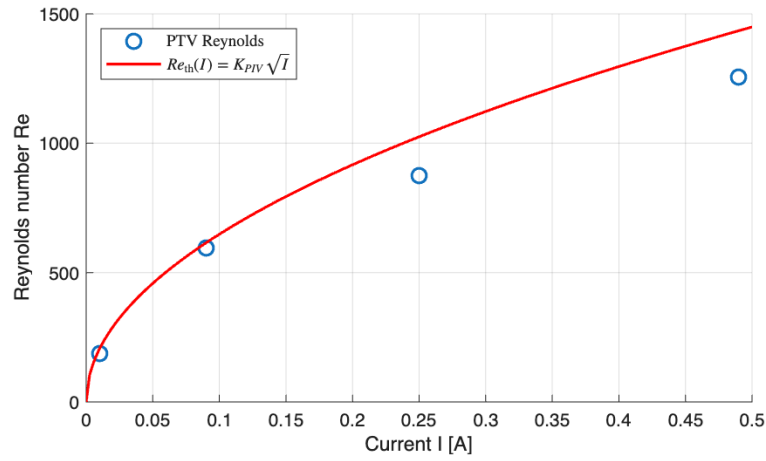


Figure 21: Evolution of the experimental Reynolds numbers compared to PIV theoretical scaling

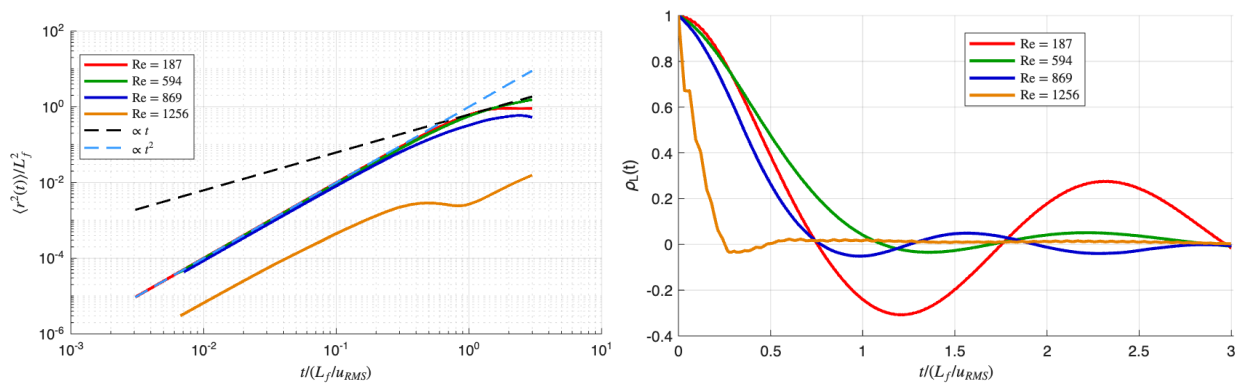


Figure 22: VACF and MSD evolution with 0.49A data